

At $x = 0$, $v = \text{maximum}$, $a = 0$.

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Let the intensity of the initial beam of light be I_0 .

Let the intensity of the beam of light after passing through the first polariser be I_1 .

Let the intensity of the beam of light passing through the second polariser be I_2 .

Given $\frac{I_2}{I_0} = \frac{1}{9}$. Since initial beam of light is unpolarised, $\frac{I_1}{I_0} = \frac{1}{2}$. Hence, $\frac{I_2}{I_1} = \frac{2}{9}$.

Applying Malus' Law, $I_2 = I_1 \cos^2 \theta$. Hence, $\frac{2}{9} I_1 = I_1 \cos^2 \theta$ and $\theta = 61.9^\circ \approx 62^\circ$ (2sf).